

Mask Detection Project: Digital Image Processing Approach

This project focuses on detecting face masks using advanced image processing. Mask detection is vital for health safety during pandemics. This presentation outlines the project's methods, technology, challenges, results, and future directions.

made by:-Rithvik Jaswal(RA2311003010040)

Akshat Gupta(RA2311003010021)



Understanding Digital Image Processing

Image Basics

Images consist of pixels and RGB color channels defining their appearance.

Preprocessing

Techniques include resizing and converting images to grayscale for simplification.

Feature Extraction

Uses Haar cascades, HOG, and CNNs to identify key image traits.

Classification

Machine learning methods like SVM and neural networks classify mask presence.

Project Overview: Mask Detection System

Problem Statement

Detecting whether faces in images wear masks accurately.

Dataset

Diverse labeled images with and without masks, sourced from public repositories.

Technologies Used

- Python programming
- OpenCV for image manipulation
- TensorFlow/Keras for modeling

Architecture

Processes include image capture, preprocessing, feature extraction, and classification.

Methodology: Feature Extraction Techniques



Haar Cascades

Detects face regions using Viola-Jones algorithm for quick localization.



HOG Features

Extracts shape and edge information from images for pattern recognition.



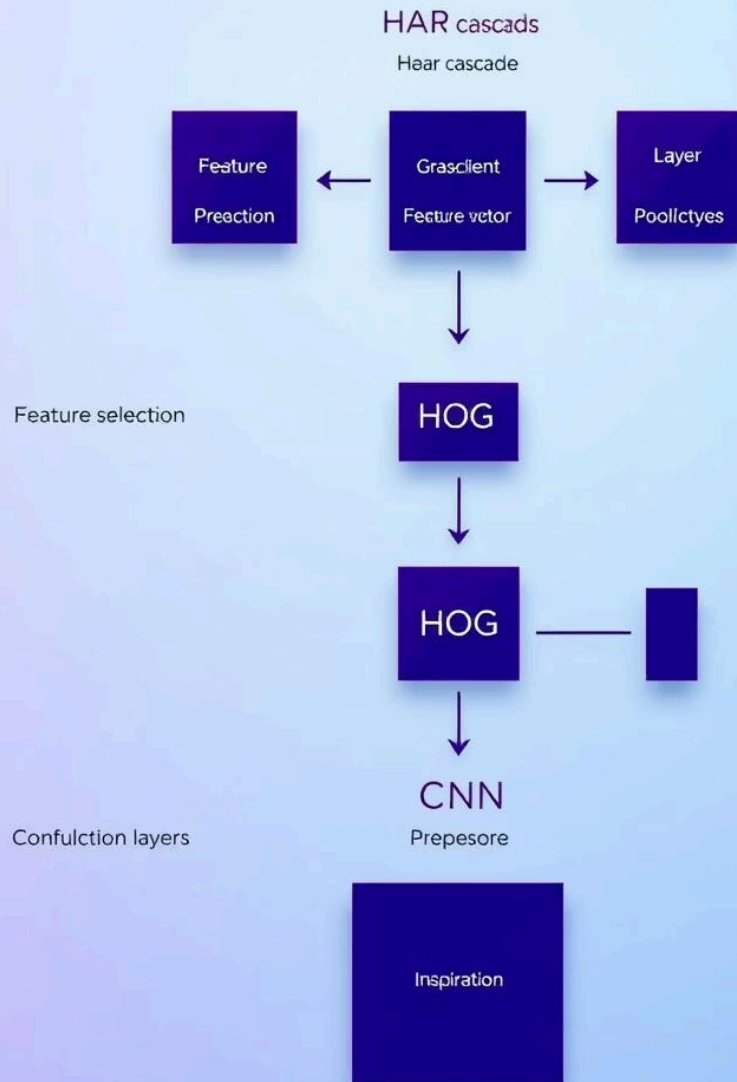
CNNs

Deep learning models that learn complex features automatically.



Comparison

Evaluates accuracy and speed differences between methods.



Model Training and Evaluation

1 CNN Architecture

Uses multiple layers with activation functions to capture features.

2 Training Setup

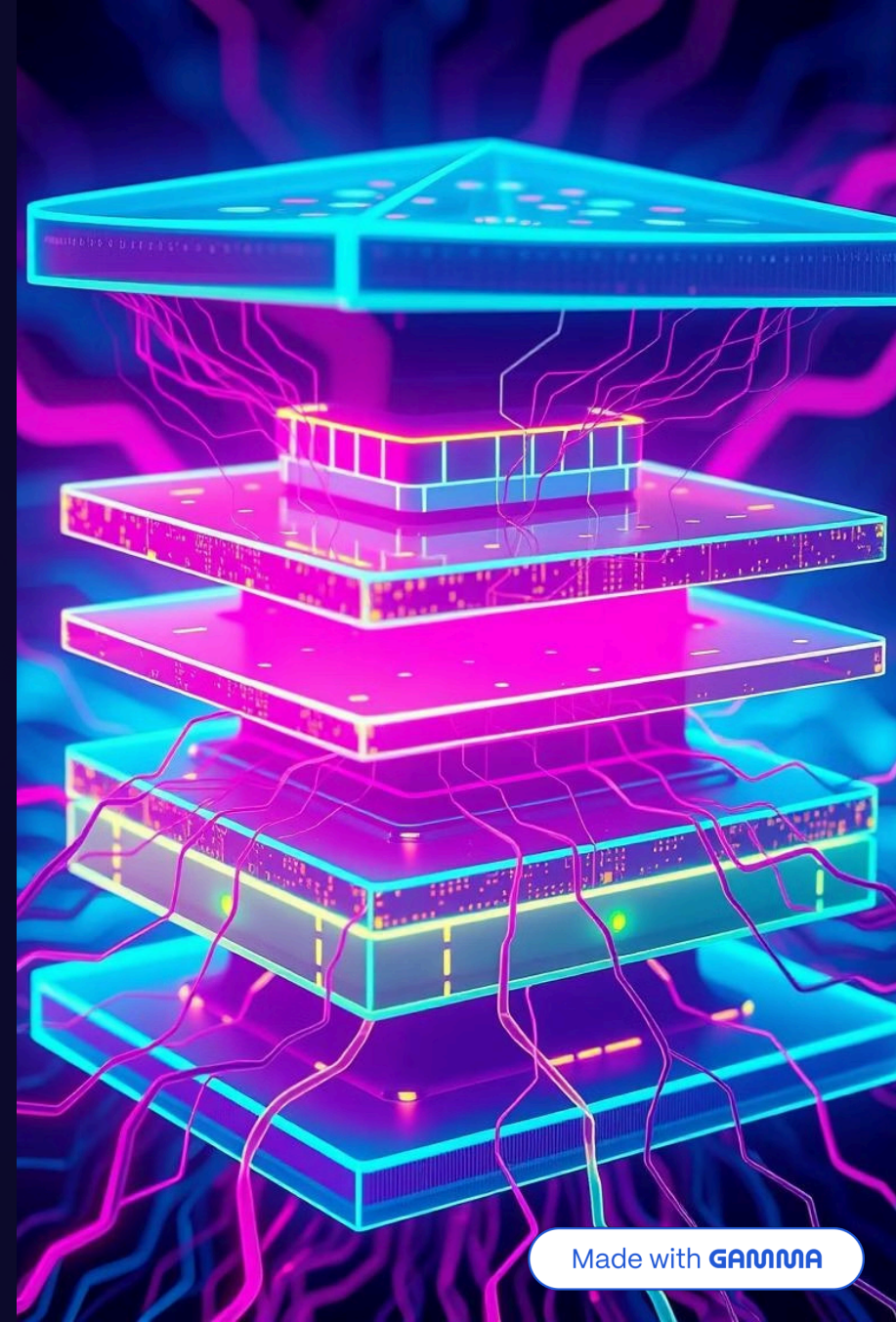
Dataset split into training and testing, tuned with epochs and batch sizes.

3 Metrics

Measures include accuracy, precision, recall, and F1-score.

4 Performance

Comparison highlights better detection of masked vs. unmasked faces.



Challenges and Solutions

Lighting Variations

Adaptive histogram equalization enhances visibility in varied lighting.

Occlusion Issues

Data augmentation generates diverse examples to improve robustness.

Real-Time Needs

Code optimization ensures fast processing for live detection.

Edge Cases

Special handling of profile views and partial masks improves accuracy.



Results and Discussion

Accuracy

The model achieves approximately 95% accuracy in mask detection.

Visual Examples

Showcases both successful detections and edge case failures.

Limitations

- Sensitivity to pose and scale variations
- Differences in detection across ethnicities

Comparison

Performance is competitive with current state-of-the-art approaches.



Conclusion and Future Improvements

1

Project Summary

Built an effective mask detection system using image processing.

2

Applications

Enhances access control and public safety in diverse settings.

3

Future Work

Focus on robustness, varied conditions, and deployment scalability.

4

Acknowledgments

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